Last Time:

- Exponential/Lie Algebra → Lie Group
- Quaternions Planar Rotations
- Geometry of quaternions

Today:

- Quaternion Algebra + Kinematics

Quaternion Algebra:

- Recall for planar rotations

$$q(\theta) = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{bmatrix}$$

- Recall axis-angle vector:

$$\phi = a\theta$$

a: unit vector, axis of rotation; θ: angle of rotation (rad)

- In 3D we generalize this to:

$$q(\phi) = \begin{bmatrix} \cos(\theta/2) \\ a\sin(\theta/2) \end{bmatrix} = \begin{bmatrix} \cos(\|\phi\|/2) \\ \frac{\phi}{\|\phi\|}\sin(\|\phi\|/2) \end{bmatrix} = \begin{bmatrix} s \\ v \end{bmatrix}$$

s: "scalar" part; v: "vector" part

* Rotations correspond to unit quaternions:

$$q^\dagger q = \cos^2(\theta/2) + \underbrace{a^\dagger a}_{1}\sin^2(\theta/2) = 1$$

$\Rightarrow$ easy to normalize (unlike rotation matrices)

* Identity Quaternion:

$$q_I = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

* Quaternion Inverse / Conjugate:

$$q^{-1} = \begin{bmatrix} \cos(-\theta/2) \\ a\sin(-\theta/2) \end{bmatrix} = \begin{bmatrix} s \\ -v \end{bmatrix} = \underbrace{q^+}_{\text{"conjugate"}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -I \end{bmatrix}}_{T} \begin{bmatrix} s \\ v \end{bmatrix} = Tq$$

("dagger")

* There are 2 quaternions for every 3D rotation:

$$\begin{bmatrix} \cos(\theta + 2\pi/2) \\ a\sin(\theta + 2\pi/2) \end{bmatrix} = \begin{bmatrix} -\cos(\theta/2) \\ -a\sin(\theta/2) \end{bmatrix} = -q$$

("double cover")

* Compose rotations by multiplying quaternions:

$$q_1 * q_2 = \begin{bmatrix} s_1 \\ v_1 \end{bmatrix} * \begin{bmatrix} s_2 \\ v_2 \end{bmatrix} = \begin{bmatrix} s_1 s_2 - v_1^T v_2 \\ s_1 v_2 + s_2 v_1 + v_1 \times v_2 \end{bmatrix}$$

$$\underbrace{\left[\begin{array}{c|c} s_1 & -v_1^T \\ \hline v_1 & s_1 I + S(v_1) \end{array}\right]}_{L(q_1)} \begin{bmatrix} s_2 \\ v_2 \end{bmatrix} = \underbrace{\left[\begin{array}{c|c} s_2 & -v_2^T \\ \hline v_2 & s_2 I - S(v_2) \end{array}\right]}_{R(q_2)} \begin{bmatrix} s_1 \\ v_1 \end{bmatrix}$$

- Note $L(q^+) = L^T(q)$, $R(q^+) = R^T(q)$

* Rotate a Vector!

$$\underbrace{\begin{bmatrix} 0 \\ {}^N x \end{bmatrix}}_{\text{"pure vector"}} = q * \begin{bmatrix} 0 \\ {}^B x \end{bmatrix} * q^{-1} \Rightarrow {}^N x = \underbrace{\begin{matrix} H^T R^T(q) L(q) H^B \\ H^T L(q) R^T(q) H^B \end{matrix}}_{{}^N_B Q} x$$

$$H = \begin{bmatrix} 0 \\ I \end{bmatrix} \Rightarrow Hx = \begin{bmatrix} 0 \\ x \end{bmatrix}$$

* Quaternions act just like rotations:

$$Q^T Q = I \longleftrightarrow q^T q = 1$$

$$Q_3 = Q_2 Q_1 \longleftrightarrow q_3 = q_2 * q_1 = L(q_2) q_1 = R(q_1) q_2$$

$$Q_1 = Q_2^T Q_3 \longleftrightarrow q_1 = q_2^+ * q_3 = L^T(q_2) q_3 = R(q_3)^T q_2$$

$$S(x) = Q S(^B x) Q^T \longleftrightarrow H {}^N x = q * H {}^B x * q^+ = L(q) R^T(q) H {}^B x = R^T(q) L(q) H {}^B x$$

- Note skew-symmetric matrix $\longleftrightarrow$ "pure vector" quaternion

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Quaternion Kinematics:

* How do we integrate a gyro? $\omega \rightarrow \dot{q} \rightarrow q(t)$

$$q_2 = q_1 * \underbrace{\Delta q}_{\text{small rotation}} = q_1 * \begin{bmatrix} \cos(\theta/2) \\ a \sin(\theta/2) \end{bmatrix} \approx q_1 * \underbrace{\begin{bmatrix} 1 \\ a\, \theta/2 \end{bmatrix}}_{\frac{1}{2}\Delta\phi \,=\, \text{small axis-angle}}$$

$$q_2 \approx q_1 * (q_I + \tfrac{1}{2} H \Delta\phi) \approx q_1 + \tfrac{1}{2} L(q_1) H \Delta\phi$$

$$\Rightarrow \frac{q_2 - q_1}{\Delta t} = \frac{\frac{1}{2} L(q_1) H \Delta\phi}{\Delta t}$$

$$\Downarrow \lim \Delta t \to 0$$

$$\dot{q} = \frac{1}{2} \underbrace{L(q) H}_{G(q)} \omega = \frac{1}{2} \underbrace{G(q)}_{4\times 3} \omega$$

"attitude Jacobian"

* Quaternion Exponential:

$$q = \exp_q(\phi) = \begin{bmatrix} \cos(\|\phi\|) \\ \frac{\phi}{\|\phi\|} \sin(\|\phi\|) \end{bmatrix}$$

- we can also define $\phi = \log_q(q)$

* Now we can simulate rigid body dynamics with quaternions!